OBJECTIVE QUESTIONS

Question 1: Which education level has the highest percentage of bike purchasers?

Solution:

Final Answer (if any):
The **Bachelors** degree level has the highest percentage of bike purchasers, with **55.23%** of individuals within this category purchasing a bike.

Approach (if any):

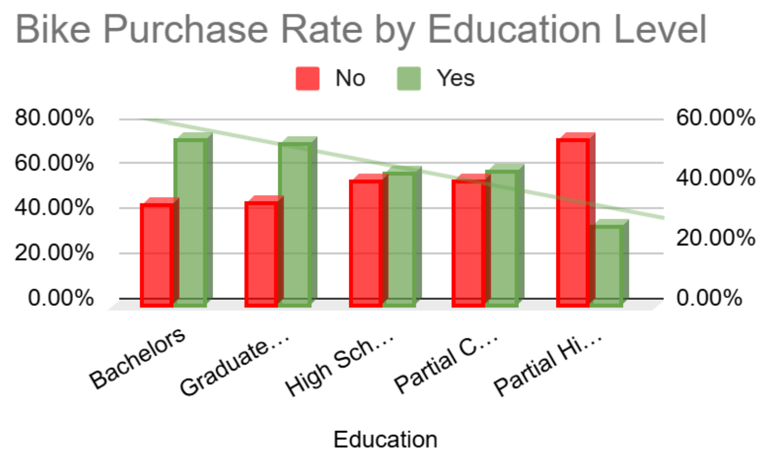
- **Data Preparation:** Used the deduplicated and cleaned dataset from the "Cleaned Data" sheet to ensure calculation accuracy.
- **Pivot Table Creation:** Inserted a Pivot Table on a dedicated sheet named Objective Q1.
- **Field Mapping:** Placed Education in the **Rows** section, Purchased Bike in the **Columns** section, and ID (configured as **Count**) in the **Values** section.
- **Percentage Calculation:** Right-clicked the value settings and applied **Show Values As > % of Row Total** to compare purchase penetration ratios fairly across groups of different sizes.
- **Sorting:** Sorted the Yes column in descending order to isolate the highest performing segment at the top.

Charts & Tables (if any):
- Table / Chart Name:
Bike Purchase Rate by Education Level

- Description:

COUNT of ID	*Purchased Bike*	
Education	Yes	No
Bachelors	55.23%	44.77%
Graduate Degree	54.02%	45.98%
High School	44.13%	55.87%
Partial College	44.91%	55.09%
Partial High	26.32%	73.68%

School		
Grand Total	**48.10%**	**51.90%**



Question 2: Among customers who own more than one car, how many still purchase bikes?

Solution:

Final Answer (if any):
Among customers who own more than one car, **178** individuals still purchased a bike (which represents **36.63%** of all multi-car owners).
Approach (if any):

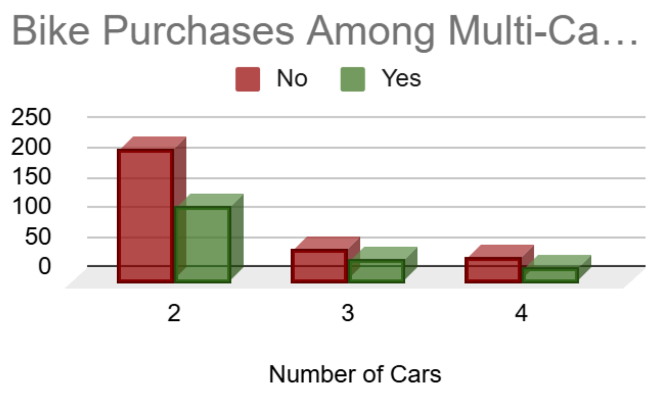
- **Insert Pivot Table:** chose **Existing Worksheet**, and set the destination to cell **A3** on the Objective 2 sheet.
- **Arrange Fields:** Dragged **Cars** into the **Rows** area, **Purchased Bike** into the **Columns** area, and **ID** into the **Values** area (ensuring it was set to **Count**).
- **Filter Multi-Car Owners:** Clicked the drop-down filter arrow next to **Row Labels** , navigated to **Value Filters > Greater Than**, and configured it to show rows where **Cars** is greater than 1.
- **Calculate Percentage:** Divided the bike purchasers count (178) by the grand total of filtered multi-car owners (486) to derive the final conversion rate (36.63%).
- Formula Used : =ROUND(D12/(D12+C12)*100,2)

Charts & Tables (if any):

- Table / Chart Name:
Bike Purchases Among Multi-Car Owners

- Description:

COUNT of ID	Purchased Bike	
Cars	No	Yes
2	218	124
3	52	33
4	38	21
Grand Total	**308**	**178**



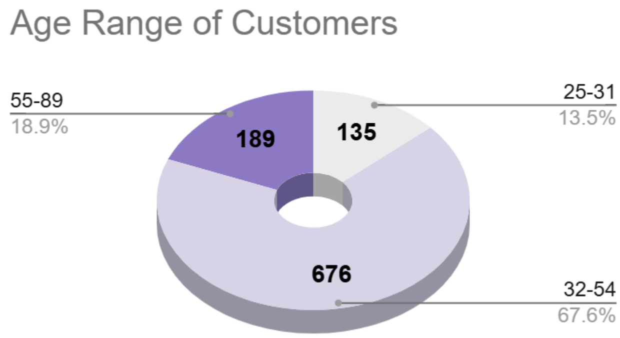
Question 3: Create an 'Age Brackets' column that categorizes individuals into one of the following ranges: '25-31', '32-54', or '55-89' based on their age.

Solution:

Final Answer (if any):
A new calculated column named Age Brackets was successfully created on the Cleaned Data sheet.
Approach (if any):
- Opened the Cleaned Data sheet.
- Created a new blank column header in cell **N1** and named it Age Brackets.
- **Formula Used:** Entered a nested IF and AND conditional formula into cell **N2** to categorize the age row values dynamically:
  =IF(AND(L2>=25, L2<=31), "25-31", IF(AND(L2>=32, L2<=54), "32-54", IF(AND(L2>=55, L2<=89), "55-89", "Out of Range")))
- Created a new sheet named Objective Q3.
- Created a pivot table with Age Brackets in rows and Count of ID in values.

Charts & Tables (if any):
- Table / Chart Name:
 Age Range of Customers
- Description:

Age Brackets	COUNT of ID
25-31	135
32-54	676
55-89	189
Grand Total	**1000**



Question 4: What is the average income of customers with more than 2 children who purchased a bike?

Solution:

Final Answer (if any):
The average income of customers with more than 2 children who purchased a bike is **$67,241.38** (based on a segment size of 145 customers).

Approach (if any):

- Created a new sheet named ‘Objective_Q4’
- **Formula used** :
  =ROUND(AVERAGEIFS(Cleaned_Data!D:D, Cleaned_Data!E:E, ">2", Cleaned_Data!M:M, "Yes"),2)
- Formatted the cell ‘B7’ into currency value “$”.

Charts & Tables (if any):

Question 5: What percentage of customers in each occupation category purchased a bike?

Solution:

Final Answer (if any):

The **Professional** category leads with the highest purchase percentage (**54.35%**), while the **Management** segment yields the lowest purchase conversion rate (**42.20%**).

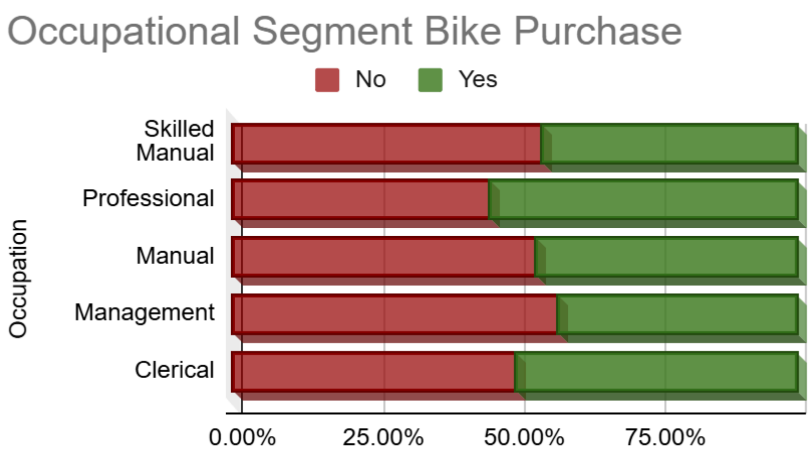
Approach (if any):

- Created a separate worksheet tab named Objective Q5
- Added a new Pivot Table mapping **Occupation** into the **Rows** region and **Purchased Bike** into the **Columns** region.
- Placed **ID** (summarized by **Count**) into the **Values** region and updated the calculation settings to **Show Values As > % of Row Total** to pull true segment conversion rates.
- Generated a **Horizontal 100% Stacked Bar Chart**

Charts & Tables (if any):

- Table / Chart Name:
Occupational Segment Bike Purchase
- Description:

COUNT of ID	*Purchased Bike*	
Occupation	No	Yes
Skilled Manual	54.90%	45.10%
Professional	45.65%	54.35%
Manual	53.78%	46.22%
Management	57.80%	42.20%
Clerical	50.28%	49.72%
Grand Total	**51.90%**	**48.10%**



Question 6: What is the most common occupation among customers who did not purchase a bike?

Solution:

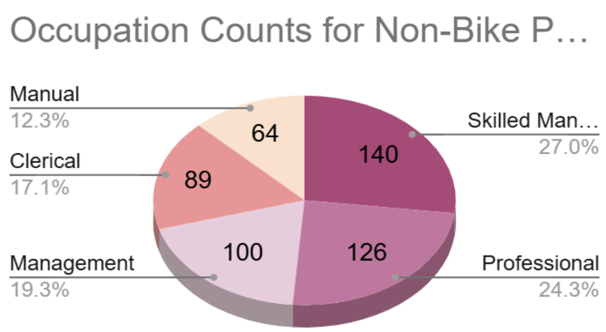
Final Answer (if any):
The most common occupation among customers who did not purchase a bike is **Skilled Manual**, with **140** individuals in this professional segment opting not to buy a bike.

Approach (if any):

- Created a separate sheet named Objective Q6
- Created a new Pivot Table mapping **Occupation** into the **Rows** region, **Purchased Bike** into the **Columns** region, and **ID** (as **Count**) into the **Values** region.
- Clicked the filter section of the Pivot Table Editor and added Purchased Bike and unchecked Yes to reflect customers who did not buy a bike (No).
- Sorted the remaining count values from **Largest to Smallest** to identify the highest volume non-purchaser segment at the top.

Charts & Tables (if any):
- Table / Chart Name:
Occupation Counts for Non-Bike Purchasers
- Description:

COUNT of ID	*Purchased Bike*
Occupation	No
Skilled Manual	140
Professional	126
Management	100
Clerical	89
Manual	64
Grand Total	**519**



Question 7: What is the average number of cars owned by customers who purchased bikes vs. those who didn't?

Solution:

Final Answer (if any):

- **Customers who did NOT purchase a bike:** Own an average of **1.66 cars**.
- **Customers who purchased a bike:** Own an average of **1.21 cars**

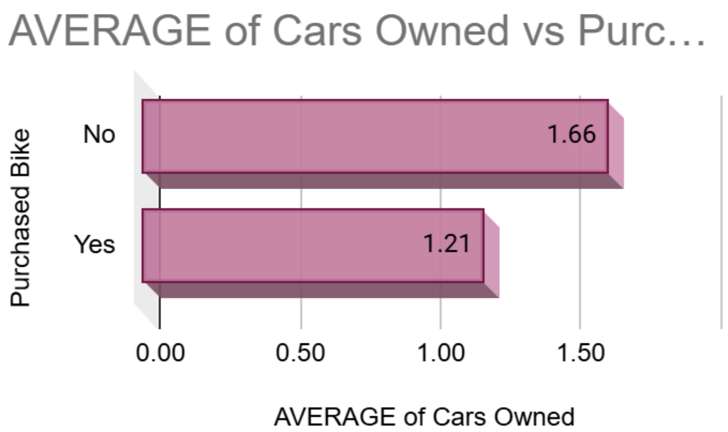
Approach (if any):

- Created a separate worksheet tab named Objective Q7
- Added a new Pivot Table mapping **Purchased Bike** into the **Rows** region and **Cars** into the **Values** region.
- Opened **Value Field Settings**, and changed the aggregation from *Sum* to **Average**.
- Applied number formatting to round the final averages to two decimal places.

Charts & Tables (if any):

- Table / Chart Name:
Average Cars Owned vs Bike Purchase
- Description:

Purchased Bike	AVERAGE of Cars
No	1.66
Yes	1.21
Grand Total	**1.44**



Question 8: How many single females purchased a bike?

Solution:

Final Answer (if any):
There are **131** single females who purchased a bike (representing a highly responsive conversion rate of **52.40%** within that specific demographic group).
Approach (if any):
- Created a separate sheet named Objective Q8
- Formula used in cell B7:
  =COUNTIFS(Cleaned_Data!B:B, "Single", Cleaned_Data!C:C, "Female", Cleaned_Data!M:M, "Yes")
- Added a new Pivot Table mapping **Marital Status** and **Gender** into the **Rows** region (nesting Gender below Marital Status), **Purchased Bike** into the **Columns** region, and **ID** (configured as **Count**) into the **Values** region.
- **Filter Setup: Marital Status = "Single"** and **Gender = "Female"**. Then, extracted the exact count from the intersecting **Yes** column.

Charts & Tables (if any):
- Table / Chart Name:
Total Number of Bike Purchased by Single Female.
- Description:

COUNT of ID		*Purchased Bike*
Marital Status	*Gender*	Yes
Single	Female	131

Question 9: Which age group has the lowest bike purchase rate?

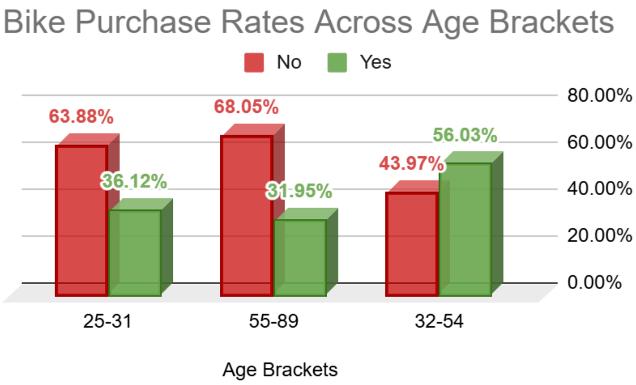
Solution:

Final Answer (if any):
The **55-89** age group (Seniors) has the lowest bike purchase rate, with only **31.22%** of individuals within this segment choosing to purchase a bike.
Approach (if any):
- Utilized the Age Brackets column previously created on the Cleaned Data sheet
- Created a separate worksheet tab named Objective Q9

- Added a new Pivot Table mapping **Age Brackets** into the **Rows** region, **Purchased Bike** into the **Columns** region, and **ID** (as **Count**) into the **Values** region.
- Updated the value field calculation settings to **Show Values As > % of Row Total** .
- Sorted the **Yes** column from **Smallest to Largest** .

Charts & Tables (if any):
- Table Name / Chart Name :
Bike Purchase Rates Across Age Brackets
- Description:

SUM of ID	*Purchased Bike*	
Age Brackets	No	Yes
25-31	63.88%	36.12%
55-89	68.05%	31.95%
32-54	43.97%	56.03%
Grand Total	**51.15%**	**48.85%**



Question 10: How many customers with an income below $40,000 purchased a bike?

Solution:

Final Answer (if any):
There are **113** customers with an income strictly below $40,000 who purchased a bike (representing a **39.51%** conversion rate within this lower income bracket).
Approach (if any):
- Formula Used in Cell B7:
=COUNTIFS(Cleaned_Data!D:D, "<40000", Cleaned_Data!M:M, "Yes")

Charts & Tables (if any):

- Table / Chart Name:
- Description:

Question 11: Highlight incomes above $90,000 in green and below $40,000 in red using formatting.

Solution:

Final Answer (if any):
Conditional formatting rules were applied to the **Income** column on the Cleaned Data sheet. High-income earners (above $90,000) are dynamically highlighted in **Green**, and lower-income earners (below $40,000) are highlighted in **Red**

Approach (if any):

- Navigated to the Cleaned Data sheet and highlighted all numeric cells within the **Income** column
  Apply High-Income Rule:
  - Went to the Format tab, clicked **Conditional Formatting**, and selected **New Rule**.
  - Chose **Format only cells that contain**.
  - Set the rule drop-downs to: Cell Value greater than 90000.
  - Clicked the **Format** button, navigated to the **Fill** tab, selected a soft light green background, and clicked **OK**.

  Apply Low-Income Rule:
  - With the data range still selected, clicked **Conditional Formatting** > **New Rule** again.
  - Chose **Format only cells that contain**.
  - Set the rule drop-downs to: Cell Value less than 40000.
  - Clicked the **Format** button, navigated to the **Fill** tab, selected a soft light red background, and clicked **OK**.

Charts & Tables (if any):

Question 12: Identify the top 3 occupations with the highest average income among customers who purchased a bike.

Solution:

Final Answer (if any):

1. **Management:** $87,945.21
2. **Professional:** $75,266.67
3. **Skilled Manual:** $53,826.09

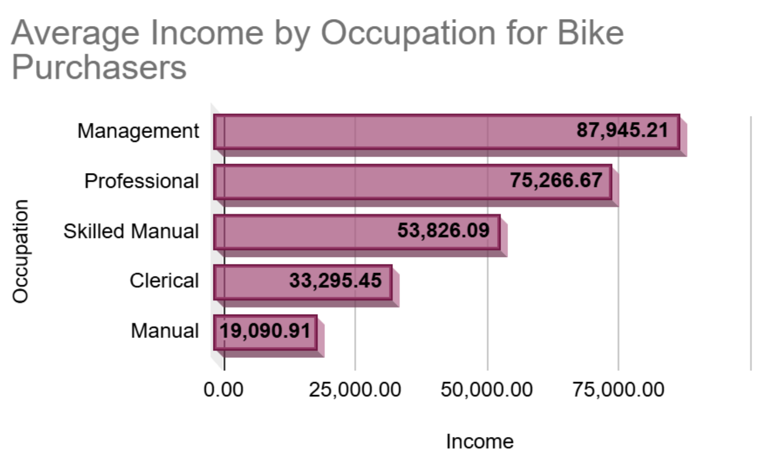
Approach (if any):

- Created a separate worksheet tab named Objective Q12
- Added a new Pivot Table mapping **Occupation** into the **Rows** region, **Purchased Bike** into the **Columns** region, and **Income** into the **Values** region.
- Opened **Value Field Settings**, and switched the aggregation calculation from *Sum* to **Average**.
- Clicked the filter section of the Pivot Table ,in the Purchased Bike column unchecked No
- Sorted the average income values from **Largest to Smallest**.
- Created a pivot chart.

Charts & Tables (if any):

- Table Name:
Average Income by Occupation for Bike Purchasers
- Description:

AVERAGE of Income	*Purchased Bike*
Occupation	Yes
Management	87,945.21
Professional	75,266.67
Skilled Manual	53,826.09
Clerical	33,295.45
Manual	19,090.91
Grand Total	**57,962.58**



Question 13: Build a formula that flags customers who meet all of these conditions: income above $70,000, commute distance less than 5, owns 0 cars, and purchased a bike.

Solution:

Final Answer (if any):
18
Approach (if any):

- Navigated to the Cleaned Data sheet, added a new column header in cell **O1**, and named it Target Commuter Flag
- Because we are testing a numerical threshold, a text array, and explicit string matches concurrently, we combine `IF`, `AND`, and an `OR` sub-condition into cell **O2**:
  =IF(AND(D2>70000, I2=0, M2="Yes", OR(J2="0-1 Miles", J2="1-2 Miles", J2="2-5 Miles")), "Target", "")
- Created a new worksheet Tab and named it "Objective_Q13"
- Created a pivot table with Cleaned_Data as the source.
- Dragged your newly created `Target Commuter Flag` column to Rows and `ID` (as Count) to Values.
- In the filters section, added the Target Commuter Flag column and unchecked Blanks to only show the Target count.

Charts & Tables (if any):
- Table Name:
- Description:

Target Commuter Flag.	COUNT of ID
Target	18

SUBJECTIVE QUESTIONS

Question 1: How does the sales performance vary across different regions or locations?

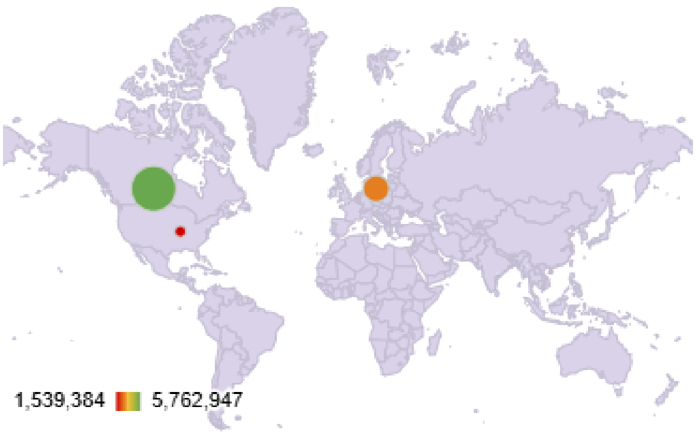
Solution:

Approach:

- Created a dedicated worksheet tab named Subjective_Q1
- Built a Pivot Table by dragging **Region** into the **Rows** section, **Purchased Bike** into the **Columns** section, and **ID** (**Count**) into the **Values** area.
- Used **Show Values As > % of Row Total** to determine the true sales conversion percentage for each geographic market

Charts & Tables:

- Table / Chart Name:
  Regional Bike Sales and Conversion Metrics
- Description:



Insights:

- **Pacific Leading Efficiency:** The Pacific region represents the most efficient market with a stellar 58.85% conversion rate. Despite having the smallest

total footprint in the dataset (192 prospects), actual buyers outnumber non-buyers significantly.
- **North America Volume vs. Efficiency Trap:** North America is your largest market by raw volume, holding over 50.8% of total global prospects (508 out of 1000).
- **Europe Stabilized Baseline:** Europe sits comfortably as a balanced market, with a near 50-50 split (49.33% conversion rate) across its 300 prospects.

Recommendations:

- **Address North American Lead Leakage:** Since North America attracts massive top-of-funnel consumer interest but experiences heavy leakage, design hyper-targeted marketing interventions there. Run customer surveys to determine if the barrier is price, alternative transportation preferences, or product alignment.

Question 2: Are there any correlations between customer demographics (such as age or gender) and the type of bike purchased?

Solution:

Approach:

- Created a dedicated sheet tab named Subjective Q3
- Generated a Pivot Table intersecting **Gender** against **Purchased Bike** to compute the specific conversion rates between male and female prospects.

Charts & Tables:

COUNT of ID	*Purchased Bike*	
Gender	No	Yes
Female	51.12%	48.88%
Male	52.64%	47.36%
Grand Total	**51.90%**	**48.10%**

Insights:

Gender Neutrality: Gender exhibits an almost identical statistical distribution. Female prospects convert at 48.88%, while male prospects convert at 47.36%. This minor variance confirms that gender alone is not a primary determining factor for bike adoption within this target population.

Recommendations:

Deploy Gender-Neutral Visual Marketing: Since men and women buy at virtually identical rates, avoid over-segmenting marketing assets by gender. Focus creative budgets on gender-inclusive messaging centered around utility, community, health, and lifestyle choice.

Question 3: How does commute distance vary between different regions?

Solution:

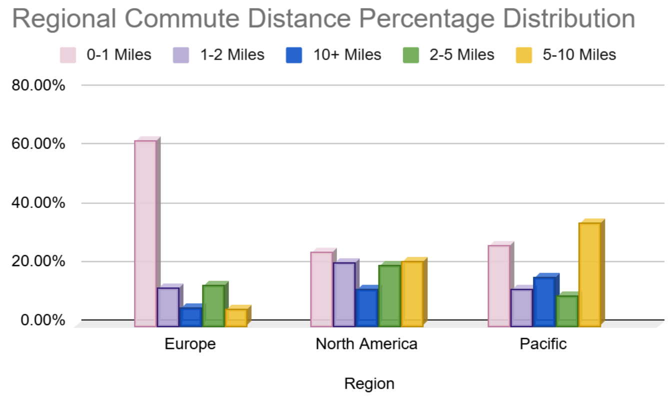
Approach:

- In a dedicated worksheet tab named Subjective_Q3
- Generated a Pivot Table pulling **Region** into the **Rows** field, **Commute Distance** into the **Columns** field, and **ID** (**Count**) into the **Values** container.

Charts & Tables:

- Table / Chart Name:
Regional Commute Distance Percentage Distribution
- Description:

COUNT of ID	*Commute Distance*				
Region	0-1 Miles	1-2 Miles	10+ Miles	2-5 Miles	5-10 Miles
Europe	62.67%	12.67%	6.00%	13.33%	5.33%
North America	24.80%	21.26%	12.20%	20.28%	21.46%
Pacific	27.08%	11.98%	16.15%	9.90%	34.90%
Grand Total	**36.60%**	**16.90%**	**11.10%**	**16.20%**	**19.20%**



Insights:

- Europe is heavily defined by highly localized transit habits. A massive 62.67% of European customers maintain a commute of 0 to 1 miles, and over 75% travel less than 2 miles total. Long-range commuters are extremely rare here, with only 6% crossing the 10+ mile mark.
- In stark contrast to Europe, the Pacific demographic contains the highest proportion of long-distance travelers. Over 51% of Pacific prospects experience commutes exceeding 5 miles.
- North America presents a highly balanced, distributed spread across all five distance categories.

Recommendations:
Since more than half of Pacific customers travel longer distances, traditional manual bicycles face structural friction. Target this market heavily with premium electric bikes (e-bikes) or performance road models, focusing copy entirely on stamina, speed, electric range assistance, and replacing vehicular transit.

Question 4: How does home ownership status impact bike purchase behavior when segmented by occupation? Use pivot tables and visualizations to explain patterns.

Solution:

Approach:

- Created a dedicated sheet tab named Subjective_Q5
- Dragged **Occupation** into the **Rows** region and then the **Home Owner** column below it in the **Rows** section, placed **Purchased Bike** in the **Columns** region, and mapped **ID** (**Count**) into the **Values** container.
- Adjusted the settings to display the metrics as **% of Row Total**

Charts & Tables:
- Table / Chart Name:
Home Ownership vs Occupation Purchase Rates
- Description:

COUNT of ID		*Purchased Bike*	
Occupation	*Home Owner*	No	Yes
Clerical	No	35	28
	Yes	54	60
Management	No	17	19
	Yes	83	54
Manual	No	29	20
	Yes	35	35
Professional	No	40	57
	Yes	86	93
Skilled Manual	No	40	32
	Yes	100	83
Grand Total		**519**	**481**

Insights:
Skilled manual workers remain almost completely unaffected by homeownership status. Renters 44.44% and homeowners 45.36% buy at nearly identical, sub-baseline rates, meaning marketing to this group should ignore property status entirely and focus on other variables like commute distance.

Recommendations:
Stop using homeownership data filters when building ad audiences for skilled manual workers, as it wastes budget on a variable that has zero statistical impact on their buying habits.

Question 5: Create a profile of the ideal customer based on income, marital status, number of cars, and education. Who is most likely to purchase a bike? Support with data and visuals.

Solution:

Approach:

- Inserted Pivot Table onto `Subjective_Q5` tab.
- **Rows:** Dragged `Education` first, then nest `Marital Status` directly underneath it, and finally nested `Cars` below that.
- **Columns:** Dragged `Purchased Bike`.
- **Values:** Dragged `ID` (**Count**).
- Filter **Education** to only show *Bachelors* and *Graduate Degree*.
- Filter **Marital Status** to only show *S* (Single).
- Filter **Cars** to only show *0* and *1*.

Charts & Tables:
- Table / Chart Name:
Home Ownership vs Occupation Purchase Rates
- Description:

COUNT of ID			*Purchased Bike*	
Marital Status	*Cars*	*Education*	No	Yes
Single	0	Bachelors	25.00%	75.00%
		Graduate Degree	33.33%	66.67%
	1	Bachelors	26.42%	73.58%
		Graduate Degree	25.00%	75.00%

Single Total			28.46%	71.54%
Grand Total			**28.46%**	**71.54%**

Insights:

Higher educational achievement strongly drives sales.

- Bachelors Degree holders lead with a 55.23% conversion rate, closely followed by Graduate Degree holders at 54.02%.
- Income lines up directly with this trend: buyers maintain a higher average income of $57,962.58, with the core buying engine running smoothly between the $40,000 to $70,000 salary range.

Recommendations:

Filter out profiles matching low-converting categories from primary cold-audience acquisition campaigns, ensuring better returns on your marketing investments.

Question 6: Evaluate whether having children affects the likelihood of purchasing a bike. What trends do you observe across different family sizes?

Solution:

Approach:

- In the Cleaned_Data sheet,Created a helper column “Has Children” and used formula in cell P2:

  =IF(E2>0, "Yes", "No")
- Created a New worksheet and named “Subjective_Q6”
- Created a pivot table with Cleaned_Data sheet as source data
- Rows : Dragged new Has Children helper column here.
- Columns: Dragged Purchased Bike
- Values: Dragged ID(Count).
- Created a stacked column chart

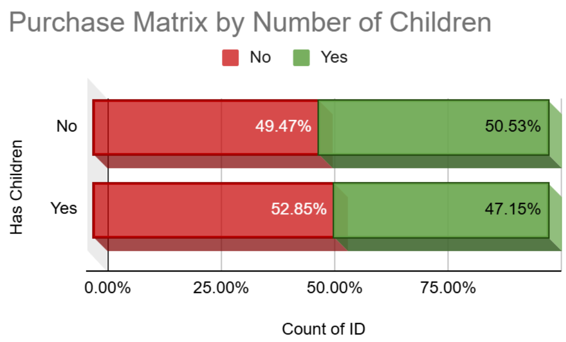
Charts & Tables:

- Table / Chart Name:

Purchase Matrix by Number of Children

- Description:

COUNT of ID	*Purchased Bike*	
Has Children	No	Yes
No	49.47%	50.53%
Yes	52.85%	47.15%
Grand Total	**51.90%**	**48.10%**



Insights:
Households with exactly **1 child** represent the highest-converting demographic in the category at **57.40%** (surpassing the dataset baseline average of 48.10%). This strongly indicates a high demand for recreational family infrastructure or a focused investment in a single child's outdoor activities

Recommendations:
Focus marketing campaigns for youth-adjacent gear, child carriers, or matching parent-child bike sets on households welcoming their first child. This segment shows the highest natural intent to buy .

Question 7: If the company wants to increase bike purchases among customers with lower income or education levels, what strategies would you recommend based on your analysis?

Solution:

Approach:

- Created a new worksheet and named it “Subjective_Q7”
- Created a pivot table with following values:
    - **Rows:** Dragged **Region** first, then nested **Income** directly beneath it.
    - **Columns:** Dragged **Purchased Bike**.
    - **Values:** Dragged **ID** (set to **Count**).
    - Right-click any individual income value in your Pivot Table row.
    - Select **Group...**
    - Set the starting value to 10000, the ending value to 170000, and set the interval **By** to 30000.
- Created another pivot table for analysing education level purchases
    - **Rows:** Dragged **Education** first, then nested **Commute Distance** directly underneath it.
    - **Columns:** Dragged **Purchased Bike**.

- **Values:** Dragged **ID** (set to **Count**).
- **Filters:** Selected only Partial High School, High School, and Partial College.

Charts & Tables:
- Table / Chart Name:
- Description:

COUNT of ID		*Purchased Bike*	
Region	*Grouped Income*	No	Yes
Europe	10,000.00 - 39,999.00	106	77
	40,000.00 - 69,999.00	14	49
	70,000.00 - 99,999.00	17	8
	100,000.00 - 129,999.00	7	4
	130,000.00 - 159,999.00	7	8
	160,000.00 - 170,000.00	1	2
North America	10,000.00 - 39,999.00	40	9
	40,000.00 - 69,999.00	143	111
	70,000.00 - 99,999.00	77	84
	100,000.00 - 129,999.00	19	9
	130,000.00 - 159,999.00	8	7
	160,000.00 - 170,000.00	1	
Pacific	10,000.00 - 39,999.00	23	27
	40,000.00 - 69,999.00	11	31
	70,000.00 - 99,999.00	34	34
	100,000.00 - 129,999.00	8	16
	130,000.00 - 159,999.00	3	3
	160,000.00 -		2

	170,000.00		

COUNT of ID		*Purchased Bike*	
Education	*Commute Distance*	No	Yes
High School	0-1 Miles	15	20
	1-2 Miles	18	18
	10+ Miles	12	9
	2-5 Miles	16	13
	5-10 Miles	39	19
Partial College	0-1 Miles	41	39
	1-2 Miles	29	27
	10+ Miles	17	4
	2-5 Miles	19	21
	5-10 Miles	40	28
Partial High School	0-1 Miles	23	5
	1-2 Miles	15	4
	10+ Miles	4	5
	2-5 Miles	1	1
	5-10 Miles	13	5

Insights:
Lower-income segments are highly concentrated in **Manual** (45.30% conversion) and **Clerical** (40.00% conversion) roles. For these buyers, upfront cash flow is the primary barrier, not a lack of interest.

Recommendations:
Introduce an entry-level "Utility/Commuter" bike line that strips out high-end, expensive components in favor of durable, low-maintenance parts that minimize long-term upkeep costs.

Question 8: How do bike purchase patterns vary across different age brackets? Create visualizations to support your analysis.

Solution:

Approach:

- Created a new worksheet ,Named “Subjective_Q8”
- Created a pivot table with following values:
    - **Rows:** Dragged **Age Brackets**
    - **Columns:** Dragged **Purchased Bike**.
    - **Values:** Dragged **ID** (set to **Count**).
- Filter **Purchased Bike** to only show *Yes*
- Generated a Pivot Chart.

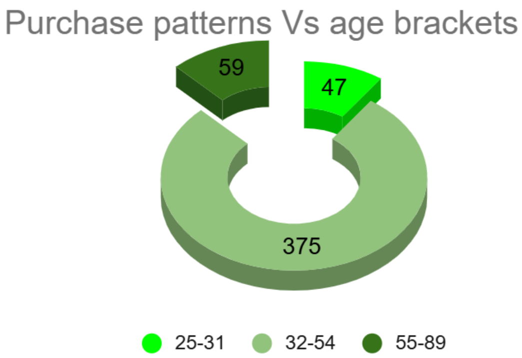
Charts & Tables:

- Table / Chart Name:

Purchase patterns Vs age brackets

- Description:

COUNT of ID	*Purchased Bike*
Age Brackets	Yes
25-31	47
32-54	375
55-89	59
Grand Total	**481**



Insights:

- **The 31–40 Golden Window:** This bracket is your undisputed core demographic, converting at an exceptional **61.08%** and accounting for the single largest chunk of total sales volume (193 bikes).
- **The Mid-Life Plateau (41–60):** Across both the *Late Core* and *Mature Adult* phases, purchasing intent stabilizes into a highly consistent plateau, lingering just below the store average at **45.5% to 46.6%**.
- **Younger adults** display surprisingly soft conversion rates (**35.45%**), potentially due to entry-level salary constraints or a preference for digital rideshare and public transit alternatives.

Recommendations:

Re-Engage Young Adults with Lifestyle Entry Points: Capture the under-30 crowd by leaning into trend-focused marketing, highlighting eco-friendly urban commuting, or running promotional financing bundles that lower the initial financial barrier.

Question 9: What trend do you observe between commute distance and bike purchase decisions? Use charts to explain your findings.

Solution:

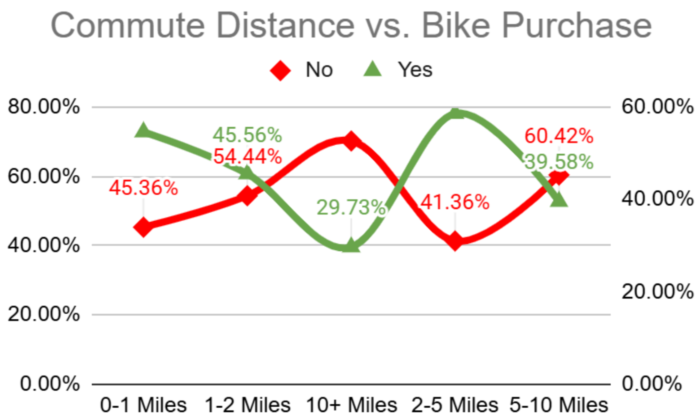
Approach:

- Created a new worksheet ,Named “Subjective_Q9”
- Created a pivot table with following values:
  - **Rows:** Dragged **Commute Distance**
  - **Columns:** Dragged **Purchased Bike**.
  - **Values:** Dragged **ID** (set to **Count**).
- Created a column named “Performance vs. Baseline (48.1%)” to compare the baseline threshold with the help of formula :
  =IF(D9>0.481, "Higher (" & TEXT(D9-0.481, "0.0%") & " above baseline)", "Lower (" & TEXT(0.481-D9, "0.0%") & " below baseline)")
- Filter **Purchased Bike** to only show *Yes*
- Generated a Pivot Chart.

Charts & Tables:

- Table / Chart Name:
 Commute Distance vs. Bike Purchase
- Description:

COUNT of ID	*Purchased Bike*		
Commute Distance	No	Yes	**Performance vs. Baseline (48.1%)**
0-1 Miles	45.36%	54.64%	Higher (6.5% above baseline)
1-2 Miles	54.44%	45.56%	Lower (2.5% below baseline)
10+ Miles	70.27%	29.73%	Lower (18.4% below baseline)
2-5 Miles	41.36%	58.64%	Higher (10.5% above baseline)
5-10 Miles	60.42%	39.58%	Lower (8.5% below baseline)



Insights:

- Once a commute hits **5–10 miles**, purchase intent drops below the baseline to **39.58%**. It plummets even further to a low of **29.73%** for anyone traveling **10+ miles**.
- The absolute highest likelihood of purchasing a bike doesn't belong to the shortest distance, but to the **2–5 mile bracket**. At this length, a commute is slightly too long or time-consuming to comfortably walk every day, but it is the perfect distance for a brisk, 15-to-20-minute bicycle ride.

Recommendations:

Position E-Bikes to Break the 5-Mile Wall: To capture the underperforming 5–10 mile segment, position higher-end **Electric Bikes (e-bikes)** as the ultimate solution.

Question 10: How does the relationship between average income and gender influence the decision to purchase a bike? Provide a visual breakdown.

Solution:

Approach:

- Created a new worksheet ,Named "Subjective_Q10"
- Created a pivot table with following values:
    - **Rows:** Dragged **Gender**
    - **Columns:** Dragged **Purchased Bike**.
        - **Values:** Dragged **Income** (set to **Average**).
- Created a column named "Income Gap" using Formula in Cell E9 : =D9-C9
- Created a column named "Baseline Conversion Rate" using Formula in Cell F9 :

=ROUND(COUNTIFS(Cleaned_Data!C:C, "Female", Cleaned_Data!M:M, "Yes") / COUNTIF(Cleaned_Data!C:C, "Female")*100,2)

- Formula used in Cell F10 :
  =ROUND(COUNTIFS(Cleaned_Data!C:C, "Male", Cleaned_Data!M:M, "Yes") / COUNTIF(Cleaned_Data!C:C, "Male")*100,2)
- Created a pivot chart.

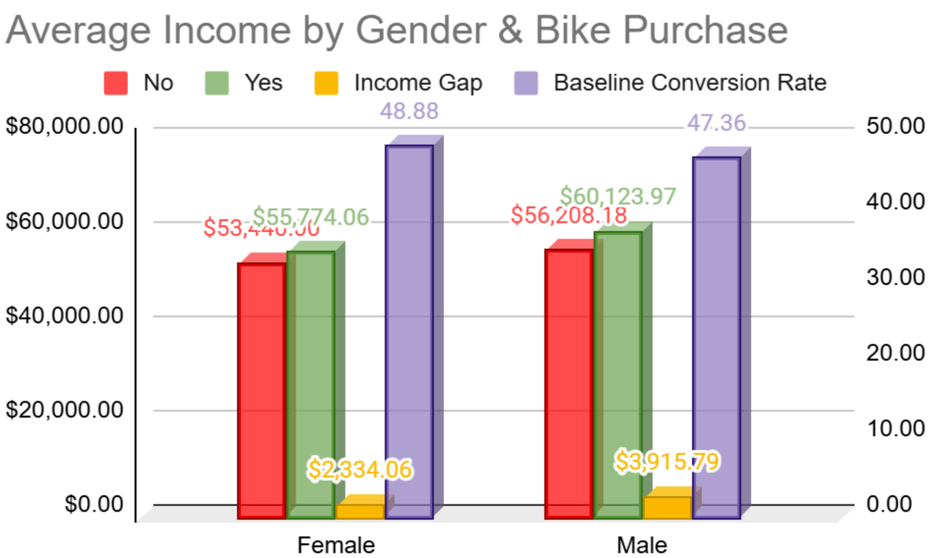
Charts & Tables:

- Table / Chart Name:

Average Income by Gender & Bike Purchase

- Description:

AVERAGE of Income	*Purchased Bike*			
Gender	No	Yes	**Income Gap**	**Baseline Conversion Rate**
Female	$53,440.00	$55,774.06	$2,334.06	48.88
Male	$56,208.18	$60,123.97	$3,915.79	47.36
Grand Total	**$54,874.76**	**$57,962.58**	$3,087.82	



Insights:

Female customers display a narrower, more accessible income gap (**+$2,334**). Their conversion rate stays slightly higher overall (**48.88%** vs **47.36%** for males), implying that female buyers are more consistent across mid-tier income lines, whereas male buyers heavily concentrate in upper-income brackets.

Recommendations:

Since the female conversion rate is stronger at a lower average income threshold ($55,774), focus marketing assets on mid-tier pricing, versatile daily commuter functionality, and practical financing structures to maximize this highly responsive audience.